

Final Project Proposal

1. The Team



Responsibilities:

- Poster: Eric Jiang, Ruibin Lyu
- Presentation: Abbie Chong, Eric Jiang
- Report: Jiping (Leo) Li, Jade Gregory
- Preparing Data: Jade Gregory, Joshua Lim, Jiping (Leo) Li

2. Problem Statement and Background

Problem Statement:

Which factors most affect the overall team score

Background:

When creating a team, the selection committees of each country have a lot to consider. A team for a country is composed of 5 gymnasts. Each major international competition has four rounds of competition: qualification, team final, all around final, and individual event finals. The last three rounds of competitions award medals. What kinds of strengths in a gymnast should each committee look for? How many all-arounders or specialists should they take? Should they take gymnasts who score better domestically or internationally? Are certain event scores correlated?

3. The Data

Link to the gymnastics dataset:

<https://drive.google.com/drive/folders/1Yg-b7hXkHbTl6SiVq1gQMtuJ56dj89Zp?usp=sharing>

Descriptive statistics:

Our scores for each competitor per apparatus ranges from 0.05 to 16.166 with the average score being 12.73006. In our data, 107 unique countries are represented across 39 unique competitions. There are 9302 female identifying competitors and 15132 male identifying competitors.

Potential data management issues:

Inconsistencies between the two data sets (competitions included, changes in rules/regulations), given that we use both data sets

Expected cleaning:

The datasets are already very clean, with few anomalies, but we do expect to pivot-wider the dataset. We also have to go in and delete some rows as the qualification scores are sometimes duplicated (teamqual, allaroundqual, and Qual). We may have to deal with the vault scores too as some gymnasts perform two vaults in order to advance to vault finals.

4. Analysis

Hypotheses:

Which events contribute most to a team score?

Do gymnastics competing in their home country tend to score higher than if they competed in a non-domestic competition?

Is there a relationship between the different event scores? Are some events stronger correlated than others?

Statistical Methods:

Correlation

2 sample t-test

ANOVA